

ELSA: Acoustic Event-Level Semantic Alignment for Fine-Grained Reference-Free Text-to-Audio Evaluation

Shuntaro Suzuki^{1,*}, Kento Tokura^{1,*}, Daichi Yashima^{1,*}, Kanon Amemiya^{1,*}

¹ Keio University, Japan

{shuntaro20021227, tkento1985, ydaichi1207, kanon-amemiya}@keio.jp

Abstract

Text-to-audio (TTA) generation, synthesizing audio directly from natural language, has been widely studied for its ability to capture precise user intent. To effectively advance TTA models, it is essential to reliably evaluate generated audio without relying on costly human subjective ratings, motivating the development of automatic evaluation metrics that correlate well with human judgments. While recent CLAP-based metrics provide practical reference-free solutions, their coarse-grained text–audio similarity matching often correlates poorly with human ratings. To address this, we propose ELSA, a reference-free evaluation metric for fine-grained text–audio alignment. ELSA decomposes generated audio guided by semantically distinct acoustic events derived from the text query and assesses event-level alignment. Experiments across four TTA benchmarks show that ELSA consistently reveals a higher correlation with human subjective ratings than prior metrics, highlighting the effectiveness of event-wise alignment for reliable TTA evaluation.¹

Index Terms: text-to-audio, automatic evaluation metric, fine-grained semantic similarity, CLAPScore

1. Introduction

Audio generation conditioned on user intent, encompassing speech, sound effects, and music, has been widely studied [1, 2]. This interest is driven by its broad range of applications, such as constructing audio environments for augmented and virtual reality [3] and synthesizing background music and sound effects for media content [4]. Among these approaches, text-to-audio (TTA) generation, which synthesizes audio directly from arbitrary text descriptions, has attracted considerable attention due to its capability of precisely reflecting detailed user intent. [5, 6].

Reliable evaluation of generated audio is essential for the effective development and benchmarking of TTA models. Human subjective evaluation of both overall audio quality (OVL) and relevance to the input text (REL) is widely regarded as the gold standard [5]. However, despite its high reliability, human evaluation is inherently costly and time-consuming, which has motivated research into automatic evaluation metrics that correlate with human subjective ratings [7, 8].

Automatic evaluation metrics for TTA can be broadly categorized into reference-based metrics [9, 8] and reference-free metrics [7, 10], depending on whether paired reference audio is required. Reference-based metrics enable fine-grained evaluation by directly comparing audio signals. For example, AudioBERTScore [8] computes frame-level similarity in

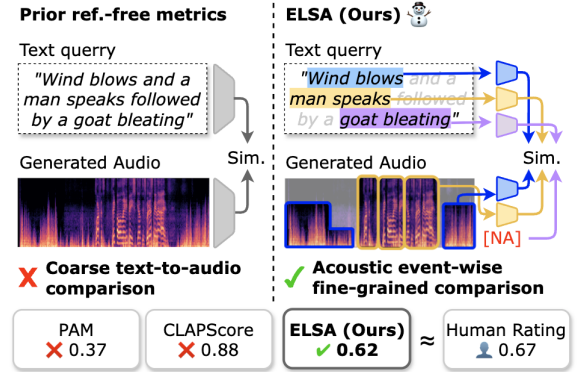


Figure 1: Overview of ELSA. ELSA enables acoustic event-wise, fine-grained text–audio alignment for reference-free TTA evaluation, yielding high correlation with human subjective ratings.

the feature space of an audio foundation model, thereby enabling implicit assessment at the level of individual acoustic events. However, the requirement for reference audio substantially limits their applicability in real-world scenarios. In contrast, reference-free metrics compare the input text with the generated audio directly, making them applicable to a wider range of experimental settings, including real-world scenarios. Nevertheless, because these methods compare fundamentally different modalities, text and audio, their evaluations are typically restricted to coarse-grained similarity measured in a jointly aligned text–audio embedding space, as exemplified by CLAP-based metrics [10]. As a result, their correlation with human subjective ratings remains limited. For instance, on the RELATE dataset, the Spearman correlation between CLAPScore and human subjective ratings of REL is only 0.280 [11].

To this end, we propose ELSA², a reference-free evaluation metric for TTA generation that enables fine-grained text–audio comparison. Figure 1 presents the core idea of the proposed approach. Unlike existing methods that rely on global text–audio matching, ELSA decomposes the text query into semantically distinct acoustic events and performs event-wise alignment with corresponding audio segments. This design allows for a more precise assessment of transient acoustic events (e.g., footsteps, door slams, or gunshots) that are often overlooked by coarse-grained metrics [10]. Experimental results show that ELSA consistently achieves higher correlation with human subjective ratings than existing metrics across four TTA benchmarks. In addition, ablation and sensitivity analyses demonstrate the contribution of individual components and the robustness of ELSA with respect to the number of target acoustic events.

^{*}These authors contributed equally.

¹Project page: <https://elsa-h0jky.kinsta.page/>

²Acoustic Event-Level Semantic Alignment score.

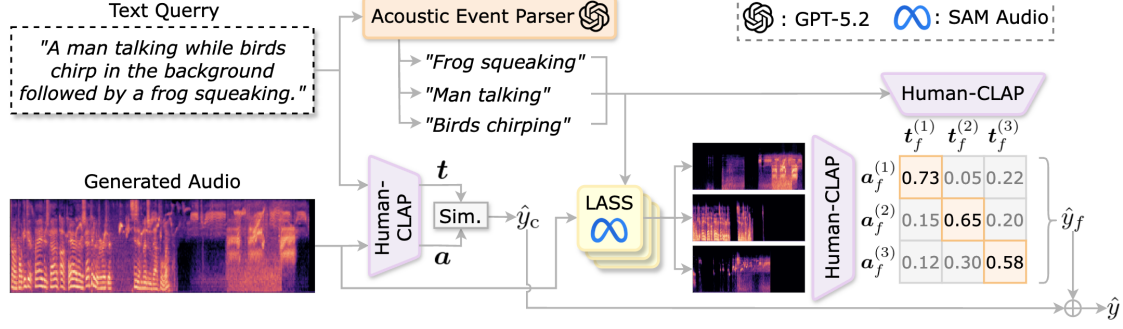


Figure 2: Architecture of ELSA. ELSA hierarchically evaluates global text–audio matching and fine-grained acoustic-event alignment by combining shared text–audio embeddings with event-level representations extracted via a text parser and a language-queried audio source separation (LASS) model.

2. Related Works

TTA generation has been extensively studied [5, 12], and its evaluation metrics have been comprehensively reviewed by Lan et al. [13] and Su et al. [14]. Although human subjective ratings remain the gold standard, their high cost has motivated the use of automatic objective metrics, particularly reference-based approaches such as SDR [15], SI-SDR [9], and feature-space distance measures computed on pretrained audio encoders, including Fréchet distance in VGGish [16] or OpenL3 [17] and KL divergence in the PaSST feature space [18]. However, these metrics rely on paired reference audio and operate solely on audio signals, which limits their applicability and precludes direct evaluation of semantic consistency between text queries and generated audio. Reference-free metrics such as PAM [7] and CLAPScore [10] address this limitation by enabling direct text–audio comparison within a shared text–audio embedding space learned by CLAP [19]. Nevertheless, because such representations are trained via instance-level contrastive learning, naive similarity matching in this space is insufficient for capturing compositional user intent [20].

3. Method

We propose ELSA, a reference-free automatic evaluation metric for TTA generation that enables fine-grained comparison between text and audio. An overview of the proposed metric is shown in Fig. 2. ELSA is inspired by prior evaluation metrics for video and image captioning [21, 22], particularly those that hierarchically assess visual–language alignment [23, 24].

3.1. Event-Level Representation Extraction

In the proposed metric, fine-grained text–audio comparison is achieved by hierarchically extracting user intent from the text query, as well as various acoustic characteristics present in generated audio. Given a text query $\mathbf{x} \in \mathbb{R}^{V \times N}$ and the corresponding generated audio $\mathbf{s} \in \mathbb{R}^L$, we first embed both modalities into a shared text–audio feature space using a pretrained embedder. This yields global text and audio representations, $\mathbf{t} \in \mathbb{R}^d$ and $\mathbf{a} \in \mathbb{R}^d$, respectively. Here, V , N , L , and d denote the vocabulary size, maximum token length, audio sequence length, and embedding dimension, respectively. Throughout our framework, we employ Human-CLAP [11]³ as the embedder, as it provides a perceptually aligned feature space reported

to correlate well with human subjective ratings. These global representations capture coarse-grained user intent reflected in both modalities. However, because the global embeddings aggregate semantics over the entire sequence, short and transient acoustic events (e.g., footsteps, door slams) can contribute less to the representation and may be obscured.

To address this limitation and better reflect fine-grained user intent specified in the text query, we additionally extract acoustic event-level representations. Specifically, we decompose the text query $\mathbf{x}$ into a set of semantically distinct events $\{\mathbf{x}_f^{(i)}\}_{i=1}^M$ using a text-only LLM. For each event description $\mathbf{x}_f^{(i)}$, we obtain a corresponding text representation $\mathbf{t}_f^{(i)}$. Furthermore, conditioning on $\mathbf{x}_f^{(i)}$, we employ a Language-queried Audio Source Separation (LASS) model to identify an event-relevant audio segment and extract its audio representation $\mathbf{a}_f^{(i)}$. In our implementation, we use GPT-5.2⁴ as the LLM and SAM-Audio [25]⁵ as the LASS model. To ensure compatibility with the LASS training distribution, each $\mathbf{x}_f^{(i)}$ is constrained to a concise noun–verb phrase (e.g., “dog barking”).

3.2. Hierarchical Alignment Scoring

Given the hierarchical acoustic event representations, we compute the final score by jointly evaluating global text–audio matching and fine-grained alignment of event-level user intent. We first compute a coarse global matching score $\hat{y}_c$ as the cosine similarity over the entire input:

$$\hat{y}_c = \frac{\mathbf{t}^\top \mathbf{a}}{\|\mathbf{t}\| \|\mathbf{a}\|}. \quad (1)$$

To further evaluate fine-grained alignment at the acoustic-event level, we define the pairwise similarity between event-level text and audio representations for each pair (i, j) as $f_{ij} = (\mathbf{t}_f^{(i)})^\top \mathbf{a}_f^{(j)} / (\|\mathbf{t}_f^{(i)}\| \|\mathbf{a}_f^{(j)}\|)$. Based on these similarities, we obtain event-level precision P_f and recall R_f as

$$P_f = \frac{1}{M} \sum_{i \in M} \left(\max_{j \in M} f_{ji} - \frac{1}{M} \sum_{j \in M} f_{ji} \right), \quad (2)$$

$$R_f = \frac{1}{M} \sum_{i \in M} \left(\max_{j \in M} f_{ij} - \frac{1}{M} \sum_{j \in M} f_{ij} \right). \quad (3)$$

³<https://github.com/sarulab-speech/Human-CLAP>

⁴<https://openai.com/index/introducing-gpt-5-2/>

⁵<https://github.com/facebookresearch/sam-audio>

Table 2: *Performance comparison on compositional benchmarks with respect to acoustic event attributes and temporal order.* “—” denotes non-executable settings.

Metrics [%]	RELATE		CompA			
	IS	OS	Attribute		Order	
	τ	τ	text $\uparrow$	audio $\uparrow$	text $\uparrow$	audio $\uparrow$
PAM	8.8	5.9	—	—	—	—
CLAPScore _{MS}	11.5	5.3	23.9	4.1	16.5	5.5
CLAPScore _{LAION}	12.9	10.2	19.3	4.1	19.0	6.0
CLAPScore _{Human}	<u>20.7</u>	10.1	17.3	3.6	<u>25.3</u>	<u>7.3</u>
ELSA (Ours)	26.7 (+6.0)	13.9 (+3.8)	23.4 (-0.5)	16.2 (+12.1)	28.0 (+2.7)	15.5 (+8.2)

Table 3: *Ablation study of the LASS model and text-audio embedding feature spaces on the Clotho benchmark.*

[%] Metrics	LASS Model	Feature Space	OVL		REL	
			ρ	τ	ρ	τ
(i)	SAM Audio	Human-CLAP	41.2	28.7	39.8	27.5
(ii)	AudioSep	Human-CLAP	40.9	28.3	43.9	30.6
(iii)	SoloAudio	Human-CLAP	37.1	25.9	36.5	25.2
(iv)	SAM Audio	MS-CLAP	28.2	19.2	26.4	17.9
(v)	SAM Audio	LAION-CLAP	31.5	21.5	30.4	20.6

beyond global similarity matching.

Next, Table 2 presents the correlation between the proposed metric and text-audio compositional similarity on the RELATE [29] and CompA [20] benchmarks. For RELATE, we evaluated the correlation on IS and OS using Kendall’s τ . For CompA, we evaluated both CompA-attribute and CompA-order tasks in terms of retrieval accuracy, considering text retrieval from audio (“text” in the table) and audio retrieval from text (“audio” in the table). As shown in Table 2, the proposed metric consistently outperformed baseline metrics on RELATE and achieved superior performance on three out of four evaluation settings on CompA. These results suggest that explicitly extracting acoustic event-level representations contributes to effective compositional comparison between text and audio.

5.2. Component Ablation Studies

To investigate the effectiveness of the proposed metric, we conducted ablation studies on its major components using the Clotho [27] benchmark. Table 3 summarizes the impact on correlation with human subjective ratings when varying the LASS model and the feature space used for text-audio similarity comparison. For the LASS model, we compared (i) SAM-Audio [25] adopted in the proposed metric with (ii) AudioSep [34] and (iii) SoloAudio [35]. For the text-audio embedding feature space, we replaced (i) Human-CLAP with (iv) MS-CLAP [19] and (v) LAION-CLAP [32].

As shown in Table 3, the performance differences induced by altering the LASS model were relatively limited: in terms of Kendall’s tau, the maximum gap was 2.8 points for OVL across Metrics (i)–(iii), and 5.4 points for REL between Metrics (ii) and (iii). In contrast, altering the feature space for text-audio comparison led to substantially larger differences, with maximum gaps of 9.5 points for OVL and 9.6 points for REL when comparing Metrics (i) and (iv). These results indicate that the proposed metric is considerably more sensitive to the choice of text-audio embedding space than to the LASS model employed.

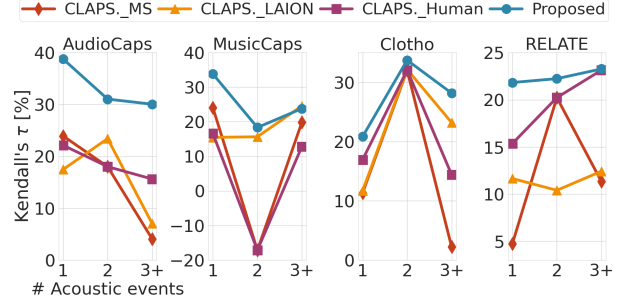


Figure 3: *Sensitivity analysis of metric-human correlation with respect to the number of acoustic events in the text query.*

5.3. Sensitivity to the Number of Acoustic Events

To examine the robustness of the proposed metric to the complexity of user intent expressed in text queries, Table 3 presents a sensitivity analysis of its correlation with human subjective ratings in terms of the number of acoustic events in each query. Here, we used the text-only LLM (GPT-5.2) to estimate the number of events per query and categorized them into three groups: one, two, and three or more events. We then compare the proposed metric with CLAPScore_{MS}, CLAPScore_{LAION}, and CLAPScore_{Human} in terms of Kendall’s τ on REL.

As shown in Table 3, the proposed metric consistently outperformed all baseline metrics across all acoustic-event conditions on four benchmarks. In particular, on the RELATE benchmark, the maximum variation across event-count conditions is 1.4 points for the proposed metric, compared to 15.5, 2.0, and 7.8 points for CLAPScore_{MS}, CLAPScore_{LAION}, and CLAPScore_{Human}, respectively. These results suggest that the proposed metric exhibits a high correlation with human subjective ratings regardless of the number of events in the text query, indicating robustness to variations in user-intent complexity.

6. Limitations

The proposed metric has two primary limitations. 1) Fine-grained text-audio similarity evaluation relies on acoustic event-level representations extracted using a LASS model, which is typically trained under the assumption that the events specified in the text query are present in the audio. In the considered task, however, such events may not be clearly manifested in the generated audio, particularly when the audio quality is low, which can reduce the reliability of the extracted representations. 2) The proposed metric does not explicitly model the temporal order of acoustic events. As shown in Table 2, ELSA nevertheless outperforms baseline metrics on order-sensitive tasks, such as OS on RELATE [29] and CompA-order on CompA [20], suggesting that temporal information is partially captured implicitly, while explicit modeling of event duration and ordering remains a promising direction for further improvement.

7. Conclusion

In this paper, we proposed ELSA, a reference-free evaluation metric for TTA generation that enables fine-grained text-audio comparison. Experimental results shows that ELSA consistently outperforms existing metrics, including both reference-based and reference-free approaches. Furthermore, sensitivity analyses indicate that ELSA is robust to complex user intent in terms of variations in the number of target acoustic events.

8. Acknowledgments

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9. Generative AI Use Disclosure

Generative AIs were used solely for auxiliary purposes, such as language refinement, manuscript formatting, and the implementation of standard algorithms. They were not involved in the research design, nor did they contribute to the development, implementation, or evaluation of the proposed methods. Accordingly, Generative AI did not constitute an essential, original, or non-standard component of this research.

10. References

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